

CPD 2021: The 4th International Workshop on Combining Physical and Data-Driven Knowledge in Ubiquitous Computing

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ABSTRACT

In the real-world ubiquitous computing systems, it is difficult to require a significant amount of data to obtain accurate information through pure data-driven methods. The performance of data-driven methods relies on the quantity and ‘quality’ of data. They perform well when a sufficient amount of data is available, which is regarded as ideal conditions. However, in real-world systems, collecting data can be costly or impossible due to practical limitations. On the other hand, it is promising to utilize physical knowledge to alleviate these issues of data limitation. The physical knowledge includes domain knowledge from experts, heuristics from experiences, analytic models of the physical phenomena and etc.

The goal of the workshop is to explore the intersection between (and the combination of) data and physical knowledge. The workshop aims to bring together domain experts that explore the physical understanding of the data, practitioners that develop systems and the researchers in traditional data-driven domains. The workshop welcomes papers, which focuses on addressing these issues in different applications/domains as well as algorithmic and systematic approaches to applying physical knowledge. Therefore, we further seek to develop a community that systematically analyzes the data quality regarding inference and evaluates the improvements from physical knowledge. Preliminary and on-going work is welcomed.

KEYWORDS

ubiquitous computing; data-driven; physical knowledge; cyber-physical system;

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1 WORKSHOP OBJECTIVE

The workshop aims to enlighten systematically discussion and development on **utilizing the physical knowledge to enhance pure data-driven methods in ubiquitous computing** as follows.

- to enable accurate learning with limited labeled data
- to explore the role of physical knowledge (including heuristics, domain knowledge, and analytic models) in computing

These objects reflect challenges for pure data-driven approaches, including

- limited available (label) data for real-world systems and applications
- spatiotemporal-varying status of the data resulting in inaccurate estimation with pure data-driven approaches.

Therefore, more and more researchers look into physical knowledge, including **heuristics, domain knowledge, analytic models**, to achieve a better understanding of the data and eventually a higher learning accuracy. However, existing work that falls into this category still lacks a systematic solution. This emerging area needs a systematic approach to enhance the now-widely-used pure data-driven approaches.

We plan to achieve this goal by bringing together researchers that facing this type of problems from different domains or working on various applications. The discussion on the limitation of the data and how to utilize the physical knowledge to enhance the ubiquitous computing performance. Various applications may face this type of data limitations in different ways. Therefore, bringing problems from different perspective together allows a broader discussion on this issue and may enable the development and generalization of these individual issues.

2 RELEVANT AND TIMELY

Ubiquitous computing community mainly focuses on human centric technologies. The data is multimodal and from heterogeneous sensing systems – from city level to person level. The challenges raise with the growth of this field of research – more papers show or discuss the limitations of pure data-driven approaches implicitly or explicitly. It is because that for the pure data-driven approaches, the data needed to understand the physical world is often insufficient to cover various conditions in real-world scenarios. As a result, more interdisciplinary works take both physical knowledge and data-driven knowledge of the data into account to achieve better inference performance.

UbiComp gathers researchers that work on various ubiquitous computing applications together, with an emphasis on design, development, and deployment. The multidisciplinary spirit of the

conference fits the topic of our workshop very well. Many applications that target to understand the human and the environment – a.k.a. the physical world – face the problem of obtaining sufficient (label) data to infer the status/information accurately.

There is no systematically approach that generalize this combination of physical and data-driven knowledge, because of the complexity of individual problems. Besides, forms of these problems vary significantly for different applications or domains. With the growth of the system studies and machine learning studies, we hope this workshop can start a new sub-field of computing that combines physical knowledge and data-driven approaches to achieve a balanced system performance for various applications.

Previous CPD workshops. Our first, second and third workshops at UbiComp 2018, 2019 and 2020 brought papers presented with explicit and implicit statements on the limitation of pure data-driven approaches appearing in different aspects. In the past three years, we received around 60 submissions from Asia, Australia, Europe and North America, which show the worldwide impact of the workshop. In addition, we attracted more than 50 audience to attend our workshop last year. Based on the historical numbers of both submissions and attendees, we expect a growing trending and higher popularity this year.

The topics/sessions we cover last year include

- system sensibility
- understand and fusing data
- using physical knowledge to enhance neural network
- human centric learning

Each drew a lot of attentions and participants had heated discussions over these topics.

3 PROPOSED ORGANIZERS AND PROGRAM COMMITTEE

The organization consists of three workshop chairs, an Advising Committee, and twelve Program Committee members. We list proposed personnel and their functions here.

3.1 Workshop Chairs

- Wenbo Ding (Tsinghua-Berkeley Shenzhen Institute)
- Chenshu Wu (University of Hong Kong)
- Weitao Xu (City University of Hong Kong)

3.2 Workshop TPC Chairs

- Xinlei Chen (Tsinghua-Berkeley Shenzhen Institute)
- Shijia Pan (University of California Merced)
- Sicong Liu (Xiamen University)

3.3 Publicity Chair

- Le Liang (Southeast University)

3.4 Web Chair

- Zihan Wang (Tsinghua-Berkeley Shenzhen Institute)

3.5 Tentative Program Committee

- Amir A. Alavi (University of Pittsburgh)
- Heba Aly (Amazon)

- Xinlei Chen (Tsinghua-Berkeley Shenzhen Institute)
- Ishara Dharmasena (Loughborough University)
- Di Jin (Amazon Alexa AI)
- Abdelwahed Khamis (University of New South Wales)
- Guohao Lan (Duke University)
- Sicong Liu (Xiamen University)
- Chengwen Luo (Shenzhen University)
- Shijia Pan (University of California Merced)
- Parth Pathak (George Mason University)
- Longfei Shangguan (Microsoft Cognition)
- Stephan Sigg (Aalto University)
- Shuai Wang (Southeast University)
- Bo Wei (Northumbria University)
- Yaxiong Xie (Princeton University)
- Susu Xu (Qualcomm AI Research)
- Fengli Xu (University of Chicago)
- Wanli Xue (University of New South Wales)
- Zimu Zhou (Singapore Management University)

The Program Committee members will conduct following tasks with the Workshop chairs:

- reviewing submitted papers (see Planned Review Procedure)
- participate review discussion if needed
- make decisions on the paper acceptance/rejection

3.6 Advising Committee

The Advising Committee members are experienced conference organizers that can provide feedback and guidance to workshop chairs.

- Pei Zhang (University of Michigan)
- Hae Young Noh (Stanford University)
- Jie Liu (Harbin Institute of Technology)
- Ercan Engin Kuruoglu (National Research Council of Italy)

3.7 Important Dates

- Submission Deadline: July 10, 2021
- Author Notifications: July 24, 2021
- Camera-ready Deadline: July 31, 2021

4 PLANNED REVIEW PROCEDURE

Each paper will be reviewed by at least three members. The criteria for accepted papers is consistent with our first workshop, including

- if the paper address or reveal the ‘limited data’ problem for pure data-driven approaches
- if the physical knowledge is used to enable better information inference
- if the paper/solution/observation is controversy or able to trigger discussion in depth regarding the combination of physical and data-driven knowledge

5 TARGET ATTENDEE AND EXPECTED NUMBERS

We target at original work on

- solving a particular problem caused by the limitation of the data through the physical knowledge of the limitation

- observation on physical phenomena that causes data limitations/inference issues in multiple applications
- introducing emerging data-driven techniques at the intersection of ubiquitous computing and machine learning

The former allows the audience to broaden their understanding of the physical knowledge that can affect inference from limited data. The later inspires more generalized approaches to be further explored. Since the limitation of the data mostly appeared in sensing data from physical systems, our target authors are mainly from (not limited to) the community of Ubicomp, CPS-IoT Week, and SenSys/BuildSys. The expected submission is 20 to 30, and the target number of papers acceptance is 12 to 20. We will host four sessions (two in the morning and two in the afternoon), and each session contains three presentations of the accepted papers. There

will be a 30 min break between each two consecutive sessions so that the audience can discuss with the author in a timely dashion.

6 PLANNED PUBLICITY PROCEDURE

We will send CFP to the relevant email lists and our research colleagues' emails. We plan to send out CFP on the following dates 05/02/2021, 05/16/2021, 05/23/2021, 05/30/2021. Other than emailing CFP, we also plan to post the workshop information on social media including Facebook, Titter, LinkedIn, and WeChat.

7 PLANNED WORKSHOP ACTIVITY

For each accepted original short paper, there will be a 12 min presentation followed by 3 min Q & A or discussion.